

The Clean Slate Architecture Checklist

12 questions to ask before rebuilding any production system. Use this framework on day one of an engagement to establish a scalable, self-healing, agent-ready stack.

01 Zero ClickOps Infrastructure

Is 100% of your cloud infrastructure defined as declarative code, with manual cloud console modifications completely disabled to prevent system drift?

TERRAFORM / OPENTOFU / AWS CLOUDFORMATION

02 Unified Monorepo Context

Are application, database, and infrastructure configurations co-located, giving development teams and AI agents absolute system context?

TURBOREPO / PNPM / GIT WORKTREE

03 Hermetic Local Sandbox

Can any new engineer spin up a complete, identical, self-contained local environment in under 15 minutes with a single terminal command?

DOCKER COMPOSE / DEVCONTAINERS / NIX

04 Strict Validation Gate

Is every pull request guarded by an automated pre-merge pipeline that blocks code integration unless lints, type checks, and full regression tests pass cleanly?

GITHUB ACTIONS / CI/CD / PRE-COMMIT

05 Continuous Observability

Do you have structured, distributed log tracing and prompt metrics wrapping every core model call and backend API entry point from day one?

LANGFUSE / GRAFANA / OPENTELEMETRY

06 Zero-Trust Secret Vaulting

Are environment credentials and API keys stored in a centralized, secure vault rather than distributed inside volatile local .env files or source code?

VAULTWARDEN / DOPPLER / CLOUD IAM

07 Declarative Schema Migrations

Are database schema modifications versioned, automated, and architected with backward compatibility to guarantee zero-downtime rolling upgrades?

PRISMA / LIQUIBASE / FLYWAY / ATLAS

08 Canary & Progressive Rollouts

Can you direct a fractional slice of production traffic (e.g., 2%) to a new version, monitoring health metrics to trigger automatic rollbacks on failure?

ARGOCD / CLOUD RUN TRAFFIC SPLIT / ISTIO

09 Absolute Environment Parity

Are development, staging, and production environments structurally identical, differing only by hardware capacity and security boundaries?

KUBERNETES / HELM / DOCKERIZATION

10 Immutable Build Artifacts

Are compilation outputs packaged into cryptographically signed, immutable containers, guaranteeing what ran in QA is exactly what runs in prod?

OCI CONTAINERS / GOOGLE ARTIFACT REGISTRY / COSIGN

11 Active Configuration Auditing

Is there an automated synchronization engine running to actively detect, alert, and auto-correct any uncoded infrastructure configuration drift?

GITOPS / ARGOCD / TERRAFORM STATE WATCHDOG

12 Agentic Readiness Standard

Is your codebase modular, documented, and declarative enough for autonomous AI coding agents to safely analyze, run, and test code sections in isolation?

SELF-DOCUMENTING CODE / SANDBOX TESTING / AI CONTEXT